第四章　積分

4.1 不定積分

定義 (反導函數)**.** 給定 $F(x)$，若 $\dfrac{\mathrm{d}}{\mathrm{d}x}F(x) = f(x)$，則稱 $F(x)$ 為 $f(x)$ 的反導函數（antiderivative）。

性質. 若 $F(x)$，$G(x)$ 分別為 $f(x)$，$g(x)$ 的反導函數，$c \in \mathbb{R}$。則

- $F(x) + c$ 為 $f(x)$ 的反導函數。
- $c\,F(x)$ 為 $c\,f(x)$ 的反導函數。
- $F(x) + G(x)$ 為 $f(x) + g(x)$ 的反導函數。

結論.

- $\dfrac{\mathrm{d}}{\mathrm{d}x}F(x) = f(x) \implies \mathrm{d}F(x) = f(x)\cdot\mathrm{d}x \implies F(x) = \displaystyle\int f(x)\cdot\mathrm{d}x = \int f(x)\,\mathrm{d}x$
- $F(x)$ 為 $f(x)$ 的反導函數 $\iff$ $f(x)$ 的反導函數為 $F(x)$ $\iff$ $F(x)$ 的導函數為 $f(x)$ $\iff$ $F(x)$（對 x）的微分為 $f(x)$ $\iff$ $f(x)$（對 x）的（不定）積分為 $F(x)$
- $f(x)$ 的反導函數 $\equiv f(x)$（對 x）的（不定）積分
- 基礎積分集：以下 $\alpha \neq -1$，$a \neq 0$。

$f(x)$	x^α	$\dfrac{1}{x}$	e^{ax}	$\sin ax$	$\cos ax$	$\dfrac{1}{\sqrt{a^2-x^2}}$	$\dfrac{1}{a^2+x^2}$
$\displaystyle\int f(x)\,\mathrm{d}x$	$\dfrac{1}{\alpha+1}x^{\alpha+1}$	$\ln\lvert x\rvert$	$\dfrac{1}{a}e^{ax}$	$-\dfrac{1}{a}\cos ax$	$\dfrac{1}{a}\sin ax$	$\sin^{-1}\dfrac{x}{\lvert a\rvert}$	$\dfrac{1}{a}\tan^{-1}\dfrac{x}{a}$

- (Liouville) e^{-x^2}，$\dfrac{e^x}{x}$，$\dfrac{1}{\ln x}$，$\sin(x^2)$，$\cos(x^2)$，$\dfrac{\sin x}{x}$，$\dfrac{\cos x}{x}$，x^x 無（初等函數形式之）反導函數！

例.

1. $\displaystyle\int \sqrt{x}\,\mathrm{d}x = \frac{2}{3}x^{\frac{3}{2}} + c$
2. $\displaystyle\int x^\pi\,\mathrm{d}x = \frac{1}{\pi+1}x^{\pi+1} + c$
3. $\displaystyle\int \sin\pi x\,\mathrm{d}x = -\frac{1}{\pi}\cos\pi x + c$
4. $\displaystyle\int \cos xy\,\mathrm{d}x = \frac{1}{y}\sin xy + c$
5. $\displaystyle\int e^{-2x}\,\mathrm{d}x = -\frac{1}{2}e^{-2x} + c$
6. $\displaystyle\int \frac{1}{\sqrt{\pi-x^2}}\,\mathrm{d}x = \sin^{-1}\frac{x}{\sqrt{\pi}} + c$
7. $\displaystyle\int \frac{1}{e+u^2}\,\mathrm{d}u = \frac{1}{\sqrt{e}}\tan^{-1}\frac{u}{\sqrt{e}} + c$
8. $\displaystyle\int \left(\frac{\pi}{x} - e^{\pi x}\right)\mathrm{d}u = \pi\ln x - \frac{e^{\pi x}}{\pi} + c$
9. $\displaystyle\int \frac{2+x^2}{1+x^2}\,\mathrm{d}u = x + 2\tan^{-1}x + c$

變數變換法

結論. $\dfrac{\mathrm{d}}{\mathrm{d}x}f(g(x)) = f'(g(x))\cdot g'(x) \implies \mathrm{d}f(g(x)) = f'(g(x))\cdot g'(x)\,\mathrm{d}x \implies f(g(x)) = \displaystyle\int f'(g(x))\cdot g'(x)\,\mathrm{d}x$。令 $u = g(x)$，則 $\dfrac{\mathrm{d}}{\mathrm{d}x}u = g'(x) \implies \mathrm{d}u = g'(x)\,\mathrm{d}x$；故 $\displaystyle\int f'(g(x))\cdot g'(x)\,\mathrm{d}x = \int f'(u)\,\mathrm{d}u = f(u) + c = f(g(x)) + c$。

例. 求 $\displaystyle\int \frac{x}{x^2+1}\,\mathrm{d}x$。

解. 令 $u = x^2+1$，則 $\mathrm{d}u = 2\,x\,\mathrm{d}x \implies x\,\mathrm{d}x = \dfrac{1}{2}\mathrm{d}u$。故 $\displaystyle\int \frac{x}{x^2+1}\,\mathrm{d}x = \int \frac{1}{x^2+1}\cdot x\,\mathrm{d}x = \int \frac{1}{u}\cdot\frac{1}{2}\,\mathrm{d}u = \frac{1}{2}\ln\lvert u\rvert + c = \frac{1}{2}\ln(x^2+1) + c$。

例. 求 $\displaystyle\int \frac{\sin(3\ln x)}{x}\,\mathrm{d}x$。

解. 令 $u=\ln x$，則 $\mathrm{d}u=\dfrac{1}{x}\,\mathrm{d}x$。故 $\displaystyle\int \frac{\sin(3\ln x)}{x}\,\mathrm{d}x=\int \sin(3\ln x)\cdot\frac{1}{x}\,\mathrm{d}x=\int \sin 3u\cdot\mathrm{d}u=-\frac{1}{3}\cos 3u+c=-\frac{1}{3}\cos(3\ln x)+c$。

例. 求 $\displaystyle\int e^x\sqrt{1+e^x}\,\mathrm{d}x$。

解. 令 $u=1+e^x$, 則 $\mathrm{d}u=e^x\,\mathrm{d}x$。故 $\displaystyle\int e^x\sqrt{1+e^x}\,\mathrm{d}x=\int \sqrt{1+e^x}\cdot e^x\,\mathrm{d}x=\int \sqrt{u}\cdot\mathrm{d}u=\frac{u^{\frac{3}{2}}}{\frac{3}{2}}+c=\frac{2}{3}(1+e^x)^{\frac{3}{2}}+c$

例. 求 $\displaystyle\int \frac{e^x}{\sqrt{2-e^{2x}}}\,\mathrm{d}x$。

解. 令 $u=e^x$，則 $\mathrm{d}u=e^x\,\mathrm{d}x$。故 $\displaystyle\int \frac{e^x}{\sqrt{2-e^{2x}}}\,\mathrm{d}x=\int \frac{1}{\sqrt{2-u^2}}\,\mathrm{d}u=\sin^{-1}\frac{u}{\sqrt{2}}+c=\sin^{-1}\frac{e^x}{\sqrt{2}}+c$。

例. 求 $\displaystyle\int \frac{1}{\sqrt{e^{2x}-1}}\,\mathrm{d}x$。

解. $\displaystyle\int \frac{1}{\sqrt{e^{2x}-1}}\,\mathrm{d}x=\int \frac{1}{e^x\sqrt{1-e^{-2x}}}\,\mathrm{d}x=\int \frac{e^{-x}}{\sqrt{1-e^{-2x}}}\,\mathrm{d}x$。令 $u=e^{-x}$，則 $\mathrm{d}u=-e^{-x}\,\mathrm{d}x \implies e^{-x}\,\mathrm{d}x=-\mathrm{d}u$；故 $\displaystyle\int \frac{e^{-x}}{\sqrt{1-e^{-2x}}}\,\mathrm{d}x=\int \frac{1}{\sqrt{1-e^{-2x}}}\cdot e^{-x}\,\mathrm{d}x=-\int \frac{1}{\sqrt{1-u^2}}\,\mathrm{d}u=-\sin^{-1}u+c=-\sin^{-1}e^{-x}+c$。

例. 求 $\displaystyle\int \frac{1}{x^2+4x+5}\,\mathrm{d}x$。

解. $\displaystyle\int \frac{1}{x^2+4x+5}\,\mathrm{d}x=\int \frac{1}{(x+2)^2+1}\,\mathrm{d}x$。令 $u=x+2$, 則 $\mathrm{d}u=\mathrm{d}x$；故 $\displaystyle\int \frac{1}{(x+2)^2+1}\,\mathrm{d}x=\int \frac{1}{u^2+1}\,\mathrm{d}u=\tan^{-1}u+c=\tan^{-1}(x+2)+c$。

例. 求 $\displaystyle\int \frac{1}{\sqrt{4+2x-x^2}}\,\mathrm{d}x$

解. $\displaystyle\int \frac{1}{\sqrt{4+2x-x^2}}\,\mathrm{d}x=\int \frac{1}{\sqrt{5-(x-1)^2}}\,\mathrm{d}x$。令 $u=x-1$，則 $\mathrm{d}u=\mathrm{d}x$；故 $\displaystyle\int \frac{1}{\sqrt{5-(x-1)^2}}\,\mathrm{d}x=\int \frac{1}{\sqrt{5-u^2}}\,\mathrm{d}u=\sin^{-1}\frac{u}{\sqrt{5}}+c=\sin^{-1}\frac{x-1}{\sqrt{5}}+c$。

例. 求 $\displaystyle\int \tan x\,\mathrm{d}x$。

解. $\displaystyle\int \tan x\,\mathrm{d}x=\int \frac{\sin x}{\cos x}\,\mathrm{d}x$。令 $u=\cos x$，則 $\mathrm{d}u=-\sin x\,\mathrm{d}x$；故 $\displaystyle\int \frac{\sin x}{\cos x}\,\mathrm{d}x=\int \frac{-1}{u}\,\mathrm{d}u=-\ln|u|+c=-\ln|\cos x|+c=\ln|\sec x|+c$

例. 求 $\displaystyle\int \sec x\,\mathrm{d}x$。

解. $\displaystyle\int \sec x\,\mathrm{d}x=\int \frac{\sec x\cdot(\sec x+\tan x)}{\sec x+\tan x}\,\mathrm{d}x$。令 $u=\sec x+\tan x$，則 $\mathrm{d}u=(\sec^2 x+\sec x\tan x)\,\mathrm{d}x$；故 $\displaystyle\int \frac{\sec x\cdot(\sec x+\tan x)}{\sec x+\tan x}\,\mathrm{d}x=\int \frac{1}{u}\,\mathrm{d}u=\ln|u|+c=\ln|\sec x+\tan x|+c$

例. 令 $T_1=\displaystyle\int\frac{\sin x}{a\cos x+b\sin x}\,\mathrm{d}x$，$T_2=\displaystyle\int\frac{\cos x}{a\cos x+b\sin x}\,\mathrm{d}x$，$a$、$b\neq 0$，求 T_1、T_2。

解.

(a) $bT_1+aT_2=\displaystyle\int\frac{b\sin x}{a\cos x+b\sin x}\,\mathrm{d}x+\int\frac{a\cos x}{a\cos x+b\sin x}\,\mathrm{d}x=\int\frac{b\sin x+a\cos x}{a\cos x+b\sin x}\,\mathrm{d}x=\int 1\,\mathrm{d}x=x$。

(b) $-aT_1+bT_2=\displaystyle\int\frac{-a\sin x}{a\cos x+b\sin x}\,\mathrm{d}x+\int\frac{b\cos x}{a\cos x+b\sin x}\,\mathrm{d}x=\int\frac{-a\sin x+b\cos x}{a\cos x+b\sin x}\,\mathrm{d}x=\int\frac{\mathrm{d}u}{u}=\ln u=\ln|a\cos x+b\sin x|$（令 $u=a\cos x+b\sin x$，則 $\mathrm{d}u=(-a\sin x+b\cos x)\,\mathrm{d}x$）。

解 T_1、T_2 方程式 (a)、(b) 得 $T_1=\dfrac{bx-a\ln|a\cos x+b\sin x|}{a^2+b^2}$，$T_2=\dfrac{ax+b\ln|a\cos x+b\sin x|}{a^2+b^2}$。

例. 求 $\displaystyle\int\cos^5 ax\,\mathrm{d}x$，$a\neq 0$。

解. $\displaystyle\int\cos^5 ax\,\mathrm{d}x=\int(1-\sin^2 ax)^2\cos ax\,\mathrm{d}x$。令 $u=\sin ax$，則 $\mathrm{d}u=a\cos ax\,\mathrm{d}x$；故 $\displaystyle\int(1-\sin^2 ax)^2\cos ax\,\mathrm{d}x=\frac{1}{a}\int(1-u^2)^2\,\mathrm{d}u=\frac{1}{a}\int(1-2u^2+u^4)\,\mathrm{d}u=\frac{1}{a}\left(u-\frac{2}{3}u^3+\frac{1}{5}u^5\right)+c=\frac{1}{a}\left(\sin ax-\frac{2}{3}\sin^3 ax+\frac{1}{5}\sin^5 ax\right)+c$。

例. 求 $\displaystyle\int\cos^4 ax\,\mathrm{d}x$，$a\neq 0$。

解. $\displaystyle\int\cos^4 ax\,\mathrm{d}x=\int\left(\frac{1+\cos 2ax}{2}\right)^2\mathrm{d}x=\frac{1}{4}\int\left(1+2\cos 2ax+\cos^2 2ax\right)\mathrm{d}x=\frac{1}{4}\int\left(1+2\cos 2ax+\frac{1+\cos 4ax}{2}\right)\mathrm{d}x$
$\displaystyle=\int\left(\frac{3}{8}+\frac{1}{2}\cos 2ax+\frac{1}{8}\cos 4ax\right)\mathrm{d}x=\frac{3}{8}x+\frac{1}{4a}\sin 2ax+\frac{1}{32a}\sin 4ax+c$

習題. 以變數變換法求下列不定積分，其中 $a\neq b\neq 0$。注意：可能會因為常數項而跟此處答案不同。

1. $\displaystyle\int e^{2x}\sin e^{2x}\,\mathrm{d}x=-\frac{1}{2}\cos e^{2x}+c$
2. $\displaystyle\int xe^{-\frac{x^2}{2}}\,\mathrm{d}x=-e^{-\frac{x^2}{2}}+c$
3. $\displaystyle\int\frac{\sin\sqrt{x}}{\sqrt{x}}\,\mathrm{d}x=-2\cos\sqrt{x}+c$
4. $\displaystyle\int x^2 2^{x^3+1}\,\mathrm{d}x=\frac{2^{x^3+1}}{3\ln 2}+c$
5. $\displaystyle\int\frac{\ln x}{x}\,\mathrm{d}x=\frac{1}{2}(\ln x)^2+c$
6. $\displaystyle\int\frac{x+1}{\sqrt{x^2+2x+3}}\,\mathrm{d}x=\sqrt{x^2+2x+3}+c$
7. $\displaystyle\int\frac{x^2}{2+x^6}\,\mathrm{d}x=\frac{1}{3\sqrt{2}}\tan^{-1}\frac{x^3}{\sqrt{2}}+c$
8. $\displaystyle\int\frac{1}{e^x+e^{-x}}\,\mathrm{d}x=\tan^{-1}e^x+c$
9. $\displaystyle\int\frac{x+1}{\sqrt{1-x^2}}\,\mathrm{d}x=-\sqrt{1-x^2}+\sin^{-1}x+c$
10. $\displaystyle\int\frac{\sec^2 x}{\sqrt{1-\tan^2 x}}\,\mathrm{d}x=\sin^{-1}(\tan x)+c$
11. $\displaystyle\int\sin^4 x\cos^5 x\,\mathrm{d}x=\frac{\sin^5 x}{5}-\frac{2\sin^7 x}{7}+\frac{\sin^9 x}{9}+c$
12. $\displaystyle\int\sin^2 x\cos^2 x\,\mathrm{d}x=\frac{x}{8}-\frac{\sin 4x}{32}+c$
13. $\displaystyle\int\sin^{-\frac{2}{3}}x\cos^3 x\,\mathrm{d}x=3\sin^{\frac{1}{3}}x-\frac{3}{7}\sin^{\frac{7}{3}}x+c$
14. $\displaystyle\int\frac{\sin^3(\ln x)\cos^3(\ln x)}{x}\,\mathrm{d}x=\frac{\sin^4(\ln x)}{4}-\frac{\sin^6(\ln x)}{6}+c$
15. $\displaystyle\int\frac{\sin^3 x}{\cos^4 x}\,\mathrm{d}x=-\frac{1}{\cos x}+\frac{1}{3\cos^3 x}+c$
16. $\displaystyle\int\cos ax\cos bx\,\mathrm{d}x=\frac{\sin(a-b)x}{2(a-b)}+\frac{\sin(a+b)x}{2(a+b)}+c$
17. $\displaystyle\int\sin ax\sin bx\,\mathrm{d}x=\frac{\sin(a-b)x}{2(a-b)}-\frac{\sin(a+b)x}{2(a+b)}+c$
18. $\displaystyle\int\sin ax\cos bx\,\mathrm{d}x=-\frac{\cos(a-b)x}{2(a-b)}-\frac{\cos(a+b)x}{2(a+b)}+c$

解.

1. 令 $u=e^{2x}$,則 $\mathrm{d}u=2e^{2x}\,\mathrm{d}x \implies e^{2x}\,\mathrm{d}x=\frac{1}{2}\,\mathrm{d}u$。故 $\int e^{2x}\sin e^{2x}\,\mathrm{d}x=\int \sin e^{2x}\cdot e^{2x}\,\mathrm{d}x=\int \sin u\cdot\frac{1}{2}\,\mathrm{d}u=-\frac{1}{2}\cos u+c=-\frac{1}{2}\cos e^{2x}+c$。

2. 令 $u=\frac{x^2}{2}$, 則 $\mathrm{d}u=x\,\mathrm{d}x$, 故 $\int xe^{-\frac{x^2}{2}}\,\mathrm{d}x=\int e^{-\frac{x^2}{2}}\cdot x\,\mathrm{d}x=\int e^{-u}\,\mathrm{d}u=-e^{-u}+c=-e^{-\frac{x^2}{2}}+c$

3. 令 $u=\sqrt{x}$,則 $\mathrm{d}u=\frac{1}{2\sqrt{x}}\,\mathrm{d}x \implies \frac{1}{\sqrt{x}}\,\mathrm{d}x=2\,\mathrm{d}u$,故 $\int \frac{\sin\sqrt{x}}{\sqrt{x}}\,\mathrm{d}x=\int \sin\sqrt{x}\cdot\frac{1}{\sqrt{x}}\,\mathrm{d}x=\int \sin u\cdot 2\,\mathrm{d}u=-2\cos u+c=-2\cos\sqrt{x}+c$

4. 令 $u=x^3+1$, 則 $\mathrm{d}u=3x^2\,\mathrm{d}x \implies x^2\,\mathrm{d}x=\frac{1}{3}\,\mathrm{d}u$, 故 $\int x^2 2^{x^3+1}\,\mathrm{d}x=\int 2^{x^3+1}\cdot x^2\,\mathrm{d}x=\int 2^u\cdot\frac{1}{3}\,\mathrm{d}u=\frac{1}{3}\int e^{u\ln 2}\,\mathrm{d}u=\frac{1}{3\ln 2}e^{u\ln 2}+c=\frac{2^{x^3+1}}{3\ln 2}+c$

5. 令 $u=\ln x$, 則 $\mathrm{d}u=\frac{1}{x}\,\mathrm{d}x$, 故 $\int \frac{\ln x}{x}\,\mathrm{d}x=\int \ln x\cdot\frac{1}{x}\,\mathrm{d}x=\int u\,\mathrm{d}u=\frac{1}{2}u^2+c=\frac{1}{2}(\ln x)^2+c$

6. 令 $u=x^2+2x+3$, 則 $\mathrm{d}u=(2x+2)\,\mathrm{d}x \implies (x+1)\,\mathrm{d}x=\frac{1}{2}\,\mathrm{d}u$, 故 $\int \frac{x+1}{\sqrt{x^2+2x+3}}\,\mathrm{d}x=\int \frac{1}{\sqrt{x^2+2x+3}}\cdot(x+1)\,\mathrm{d}x=\int \frac{1}{\sqrt{u}}\cdot\frac{1}{2}\,\mathrm{d}u=\sqrt{u}+c=\sqrt{x^2+2x+3}+c$

7. 令 $u=x^3$, 則 $\mathrm{d}u=3x^2\,\mathrm{d}x \implies x^2\,\mathrm{d}x=\frac{1}{3}\,\mathrm{d}u$, 故 $\int \frac{x^2}{2+x^6}\,\mathrm{d}x=\int \frac{1}{2+x^6}\cdot x^2\,\mathrm{d}x=\int \frac{1}{2+u^2}\cdot\frac{1}{3}\,\mathrm{d}u=\frac{1}{3\sqrt{2}}\tan^{-1}\frac{u}{\sqrt{2}}+c=\frac{1}{3\sqrt{2}}\tan^{-1}\frac{x^3}{\sqrt{2}}+c$

8. $\int \frac{1}{e^x+e^{-x}}\,\mathrm{d}x=\int \frac{e^x}{e^{2x}+1}\,\mathrm{d}x$。令 $u=e^x$, 則 $\mathrm{d}u=e^x\,\mathrm{d}x$, 故 $\int \frac{e^x}{e^{2x}+1}\,\mathrm{d}x=\int \frac{1}{e^{2x}+1}\cdot e^x\,\mathrm{d}x=\int \frac{1}{u^2+1}\,\mathrm{d}u=\tan^{-1}u+c=\tan^{-1}e^x+c$。

9. $\int \frac{x+1}{\sqrt{1-x^2}}\,\mathrm{d}x=\int \frac{x}{\sqrt{1-x^2}}\,\mathrm{d}x+\int \frac{1}{\sqrt{1-x^2}}\,\mathrm{d}x$。令 $u=\sqrt{1-x^2}$, 則 $\mathrm{d}u=\frac{-x}{\sqrt{1-x^2}}\,\mathrm{d}x \implies \frac{x}{\sqrt{1-x^2}}\,\mathrm{d}x=-\mathrm{d}u$,故 $\int \frac{x}{\sqrt{1-x^2}}\,\mathrm{d}x=\int -\mathrm{d}u=-u+c=-\sqrt{1-x^2}$, $\int \frac{x+1}{\sqrt{1-x^2}}\,\mathrm{d}x=\int \frac{x}{\sqrt{1-x^2}}\,\mathrm{d}x+\int \frac{1}{\sqrt{1-x^2}}\,\mathrm{d}x=-\sqrt{1-x^2}+\sin^{-1}x+c$。

10. 令 $u=\tan x$, 則 $\mathrm{d}u=\sec^2 x\,\mathrm{d}x$, 故 $\int \frac{\sec^2 x}{\sqrt{1-\tan^2 x}}\,\mathrm{d}x=\int \frac{1}{\sqrt{1-\tan^2 x}}\cdot\sec^2 x\,\mathrm{d}x=\int \frac{1}{\sqrt{1-u^2}}\,\mathrm{d}u=\sin^{-1}u+c=\sin^{-1}(\tan x)+c$

11. $\int \sin^4 x\cos^5 x\,\mathrm{d}x=\int \sin^4 x\cdot\cos^4 x\cdot\cos x\,\mathrm{d}x=\int \sin^4 x\cdot(1-\sin^2 x)^2\cdot\cos x\,\mathrm{d}x=\int \left(\sin^4 x-2\sin^6 x+\sin^8 x\right)\cdot\cos x\,\mathrm{d}x$。令 $u=\sin x$, 則 $\mathrm{d}u=\cos x\,\mathrm{d}x$, 故 $\int \left(\sin^4 x-2\sin^6 x+\sin^8 x\right)\cdot\cos x\,\mathrm{d}x=\int \left(u^4-2u^6+u^8\right)\mathrm{d}u=\frac{u^5}{5}-\frac{2u^7}{7}+\frac{u^9}{9}+c=\frac{\sin^5 x}{5}-\frac{2\sin^7 x}{7}+\frac{\sin^9 x}{9}+c$

12. $\int \sin^2 x\cos^2 x\,\mathrm{d}x=\int \frac{1-\cos 2x}{2}\cdot\frac{1+\cos 2x}{2}\,\mathrm{d}x=\frac{1}{4}\int \left(1-\cos^2 2x\right)\mathrm{d}x=\frac{1}{4}\int \left(1-\frac{1+\cos 4x}{2}\right)\mathrm{d}x=\frac{1}{4}\int \left(\frac{1}{2}+\frac{\cos 4x}{2}\right)\mathrm{d}x=\frac{x}{8}-\frac{\sin 4x}{32}+c$

13. $\int \sin^{-\frac{2}{3}} x\cos^3 x\,\mathrm{d}x=\int \sin^{-\frac{2}{3}} x\cdot\cos^2 x\cdot\cos x\,\mathrm{d}x=\int \sin^{-\frac{2}{3}} x\cdot(1-\sin^2 x)\cdot\cos x\,\mathrm{d}x=\int \left(\sin^{-\frac{2}{3}} x-\right.$

$\sin^{\frac{4}{3}} x\big)\cdot\cos x\,\mathrm{d}x$。令 $u=\sin x$, 則 $\mathrm{d}u=\cos x\,\mathrm{d}x$, 故 $\int\big(\sin^{-\frac{2}{3}} x-\sin^{\frac{4}{3}} x\big)\cdot\cos x\,\mathrm{d}x=\int\big(u^{-\frac{2}{3}}-u^{\frac{4}{3}}\big)\,\mathrm{d}u=3u^{\frac{1}{3}}-\dfrac{3}{7}u^{\frac{7}{3}}+c=3\sin^{\frac{1}{3}} x-\dfrac{3}{7}\sin^{\frac{7}{3}} x+c$

14. 令 $u=\ln x$,則 $\mathrm{d}u=\dfrac{1}{x}\,\mathrm{d}x$,故 $\int\dfrac{\sin^3(\ln x)\cos^3(\ln x)}{x}\,\mathrm{d}x=\int\sin^3(\ln x)\cos^3(\ln x)\cdot\dfrac{1}{x}\,\mathrm{d}x=\int\sin^3 u\cos^3 u\,\mathrm{d}u=\int\sin^3 u\cdot\cos^2 u\cdot\cos u\,\mathrm{d}u=\int\sin^3 u\big(1-\sin^2 u\big)\cdot\cos u\,\mathrm{d}u=\int\big(\sin^3 u-\sin^5 u\big)\cdot\cos u\,\mathrm{d}u$。令 $w=\sin u$, 則 $\mathrm{d}w=\cos u\,\mathrm{d}u$, 故 $\int\big(\sin^3 u-\sin^5 u\big)\cdot\cos u\,\mathrm{d}u=\int\big(w^3-w^5\big)\,\mathrm{d}w=\dfrac{w^4}{4}-\dfrac{w^6}{6}+c=\dfrac{\sin^4(\ln x)}{4}-\dfrac{\sin^6(\ln x)}{6}+c$

15. $\int\dfrac{\sin^3 x}{\cos^4 x}\,\mathrm{d}x=\int\dfrac{\sin^2 x\cdot\sin x}{\cos^4 x}\,\mathrm{d}x=\int\dfrac{\sin^2 x}{\cos^4 x}\cdot\sin x\,\mathrm{d}x=\int\dfrac{1-\cos^2 x}{\cos^4 x}\cdot\sin x\,\mathrm{d}x=\int\big(\dfrac{1}{\cos^4 x}-\dfrac{1}{\cos^2 x}\big)\cdot\sin x\,\mathrm{d}x$。令 $u=\cos x$, 則 $\mathrm{d}u=-\sin x\,\mathrm{d}x\implies\sin x\,\mathrm{d}x=-\mathrm{d}u$, 故 $\int\big(\dfrac{1}{\cos^4 x}-\dfrac{1}{\cos^2 x}\big)\cdot\sin x\,\mathrm{d}x=\int\big(\dfrac{1}{u^4}-\dfrac{1}{u^2}\big)(-)\,\mathrm{d}u=\int\big(\dfrac{1}{u^2}-\dfrac{1}{u^4}\big)\,\mathrm{d}u=-\dfrac{1}{u}+\dfrac{1}{3u^3}+c=-\dfrac{1}{\cos x}+\dfrac{1}{3\cos^3 x}+c$

16. 由積化和差公式 $\cos\alpha\cos\beta=\dfrac{1}{2}\big(\cos(\alpha-\beta)+\cos(\alpha+\beta)\big)$, $\int\cos ax\cos bx\,\mathrm{d}x=\int\dfrac{1}{2}\big(\cos(ax-bx)+\cos(ax+bx)\big)\,\mathrm{d}x=\int\dfrac{1}{2}\big(\cos(a-b)x+\cos(a+b)x\big)\,\mathrm{d}x=\dfrac{\sin(a-b)x}{2(a-b)}+\dfrac{\sin(a+b)x}{2(a+b)}+c$。

17. 由積化和差公式 $\sin\alpha\sin\beta=\dfrac{1}{2}\big(\cos(\alpha-\beta)-\cos(\alpha+\beta)\big)$, $\int\sin ax\sin bx\,\mathrm{d}x=\int\dfrac{1}{2}\big(\cos(ax-bx)-\cos(ax+bx)\big)\,\mathrm{d}x=\int\dfrac{1}{2}\big(\cos(a-b)x-\cos(a+b)x\big)\,\mathrm{d}x=\dfrac{\sin(a-b)x}{2(a-b)}-\dfrac{\sin(a+b)x}{2(a+b)}+c$。

18. 由積化和差公式 $\sin\alpha\cos\beta=\dfrac{1}{2}\big(\sin(\alpha-\beta)+\sin(\alpha+\beta)\big)$, $\int\sin ax\cos bx\,\mathrm{d}x=\int\dfrac{1}{2}\big(\sin(ax-bx)+\sin(ax+bx)\big)\,\mathrm{d}x=\int\dfrac{1}{2}\big(\sin(a-b)x+\sin(a+b)x\big)\,\mathrm{d}x=-\dfrac{\cos(a-b)x}{2(a-b)}-\dfrac{\cos(a+b)x}{2(a+b)}+c$。

部份積分法

結論. $\dfrac{\mathrm{d}}{\mathrm{d}x}\big(u(x)\,v(x)\big)=u(x)\,\dfrac{\mathrm{d}v(x)}{\mathrm{d}x}+v(x)\,\dfrac{\mathrm{d}u(x)}{\mathrm{d}x}\implies u(x)\,v(x)=\int u(x)\,\mathrm{d}v(x)+\int v(x)\,\mathrm{d}u(x)$
$\implies\int u\,\mathrm{d}v=u\,v-\int v\,\mathrm{d}u$。

例. 求 $\int x\,e^x\,\mathrm{d}x$。

解. 令 $u=x$, 則 $\mathrm{d}u=\mathrm{d}x$。令 $\mathrm{d}v=e^x\,\mathrm{d}x$, 則 $v=e^x$。故 $\int x\,e^x\,\mathrm{d}x=x\cdot e^x-\int e^x\,\mathrm{d}x=xe^x-e^x+c$。

例. 求 $\int\ln x\,\mathrm{d}x$。

解. 令 $u=\ln x$, 則 $\mathrm{d}u=\dfrac{1}{x}\mathrm{d}x$。令 $\mathrm{d}v=\mathrm{d}x$, 則 $v=x$。故 $\int\ln x\,\mathrm{d}x=\ln x\cdot x-\int x\cdot\dfrac{1}{x}\,\mathrm{d}x=x\ln x-x+c$。

例. 求 $\int x\tan^{-1} x\,\mathrm{d}x$。

解. 令 $u = \tan^{-1} x$，則 $\mathrm{d}u = \dfrac{1}{1+x^2}\mathrm{d}x$。令 $\mathrm{d}v = x\,\mathrm{d}x$，則 $v = \dfrac{x^2}{2}$。故 $\displaystyle\int x\tan^{-1} x\,\mathrm{d}x = \tan^{-1} x\cdot\frac{x^2}{2} - \frac{1}{2}\int\frac{x^2}{1+x^2}\,\mathrm{d}x = \tan^{-1} x\cdot\frac{x^2}{2} - \frac{1}{2}\int\left(1-\frac{1}{1+x^2}\right)\mathrm{d}x = \tan^{-1} x\cdot\frac{x^2}{2} - \frac{1}{2}\left(x-\tan^{-1} x\right) + c$。

例. 求 $\displaystyle\int x^2\sin x\,\mathrm{d}x$。

解. 令 $u = x^2$,則 $\mathrm{d}u = 2x\,\mathrm{d}x$;令 $\mathrm{d}v = \sin x\,\mathrm{d}x$,則 $v = -\cos x$。故 $\displaystyle\int x^2\sin x\,\mathrm{d}x = x^2\cdot(-\cos x)+2\int x\cos x\,\mathrm{d}x$。令 $u = x$, 則 $\mathrm{d}u = \mathrm{d}x$; 令 $\mathrm{d}v = \cos x\,\mathrm{d}x$, 則 $v = \sin x$。故 $\displaystyle\int x\cos x\,\mathrm{d}x = x\cdot\sin x - \int\sin x\,\mathrm{d}x = x\cdot\sin x + \cos x$。由上，$\displaystyle\int x^2\sin x\,\mathrm{d}x = -x^2\cos x + 2(x\sin x + \cos x) + c$。

例. 求 $\displaystyle\int e^{ax}\cos bx\,\mathrm{d}x$ 與 $\displaystyle\int e^{ax}\sin bx\,\mathrm{d}x$，$a$、$b \neq 0$。

解.

(a) 考慮 $\displaystyle\int e^{ax}\cos bx\,\mathrm{d}x$：令 $u = \cos bx$，則 $\mathrm{d}u = -b\sin bx\,\mathrm{d}x$；令 $\mathrm{d}v = e^{ax}\,\mathrm{d}x$，則 $v = \dfrac{1}{a}e^{ax}$，故 $\displaystyle\int e^{ax}\cos bx\,\mathrm{d}x = \cos bx\cdot\frac{1}{a}e^{ax} + \frac{b}{a}\int e^{ax}\cdot\sin bx\,\mathrm{d}x$。

(b) 考慮 $\displaystyle\int e^{ax}\sin bx\,\mathrm{d}x$:令 $u = \sin bx$,則 $\mathrm{d}u = b\cos bx\,\mathrm{d}x$;令 $\mathrm{d}v = e^{ax}\,\mathrm{d}x$,則 $v = \dfrac{1}{a}e^{ax}$,故 $\displaystyle\int e^{ax}\sin bx\,\mathrm{d}x = \sin bx\cdot\frac{1}{a}e^{ax} - \frac{b}{a}\int e^{ax}\cdot\cos bx\,\mathrm{d}x$。

令 $X = \displaystyle\int e^{ax}\cos bx\,\mathrm{d}x$，$Y = \displaystyle\int e^{ax}\sin bx\,\mathrm{d}x$，由 (a)(b) $X = \cos bx\cdot\dfrac{1}{a}e^{ax} + \dfrac{b}{a}Y$，$Y = \sin bx\cdot\dfrac{1}{a}e^{ax} - \dfrac{b}{a}X$。解 X、Y 得 $X = \dfrac{e^{ax}(a\cos bx + b\sin bx)}{a^2+b^2}$，$Y = \dfrac{e^{ax}(a\sin bx - b\cos bx)}{a^2+b^2}$。

列表法

結論.

1. 將要微分函數寫左邊、積分函數寫右邊；左邊連續微分、右邊連續積分
2. 依序左上連右下斜線函數相乘，最底部水平兩邊函數相乘並積分，符號正負相間
3. 將上式所得項全部加總即為所求積分

例. 求 $\displaystyle\int(x+3)\,e^{2x}\,\mathrm{d}x$。

解.

Diff		Int
$x+3$	$+$	e^{2x}
1	$-$	$\frac{1}{2}e^{2x}$
0		$\frac{1}{4}e^{2x}$

$$\int(x+3)\,e^{2x}\,\mathrm{d}x = (x+3)\,\frac{1}{2}e^{2x} - \frac{1}{4}e^{2x}$$

例. 求 $\displaystyle\int(x^2-2x)\,e^{kx}\,\mathrm{d}x$。

解.

Diff		Int
x^2-2x	$+$	e^{kx}
$2(x-1)$	$-$	$\frac{1}{k}e^{kx}$
2	$+$	$\frac{1}{k^2}e^{kx}$
0		$\frac{1}{k^3}e^{kx}$

$$\int (x^2-2x)\,e^{kx}\,\mathrm{d}x = (x^2-2x)\,\frac{1}{k}e^{kx} - 2(x-1)\frac{1}{k^2}e^{kx} + 2\frac{1}{k^3}e^{kx}$$

例. 求 $\int x^4\,\sin 2x\,\mathrm{d}x$。

解.

Diff		Int
x^4	$+$	$\sin 2x$
$4x^3$	$-$	$-\frac{1}{2}\cos 2x$
$12x^2$	$+$	$-\frac{1}{4}\sin 2x$
$24x$	$-$	$\frac{1}{8}\cos 2x$
24	$+$	$\frac{1}{16}\sin 2x$
0		$-\frac{1}{32}\cos 2x$

$$\begin{aligned}
&\int x^4\,\sin 2x\,\mathrm{d}x \\
&= -x^4\,\frac{1}{2}\cos 2x + 4x^3\,\frac{1}{4}\sin 2x + 12x^2\,\frac{1}{8}\cos 2x - 24x\,\frac{1}{16}\sin 2x - 24\,\frac{1}{32}\cos 2x \\
&= \left(-\frac{x^4}{2}+\frac{3x^2}{2}-\frac{3}{4}\right)\cos 2x + \left(x^3-\frac{3x}{2}\right)\sin 2x
\end{aligned}$$

例. 求 $\int x^5\,e^{ax}\,\mathrm{d}x$。

解.

Diff		Int
x^5	$+$	e^{ax}
$5x^4$	$-$	$\frac{1}{a}e^{ax}$
$20x^3$	$+$	$\frac{1}{a^2}e^{ax}$
$60x^2$	$-$	$\frac{1}{a^3}e^{ax}$
$120x$	$+$	$\frac{1}{a^4}e^{ax}$
120	$-$	$\frac{1}{a^5}e^{ax}$
0		$\frac{1}{a^6}e^{ax}$

$$\begin{aligned}
&\int x^5\,e^{ax}\,\mathrm{d}x \\
&= x^5\,\frac{1}{a}e^{ax} - 5x^4\,\frac{1}{a^2}e^{ax} + 20x^3\,\frac{1}{a^3}e^{ax} - 60x^2\,\frac{1}{a^4}e^{ax} + 120x\,\frac{1}{a^5}e^{ax} - 120\,\frac{1}{a^6}e^{ax} \\
&= \left(\frac{x^5}{a}-\frac{5x^4}{a^2}+\frac{20x^3}{a^3}-\frac{60x^2}{a^4}+\frac{120x}{a^5}-\frac{120}{a^6}\right)e^{ax}
\end{aligned}$$

例. 求 $\int (\sin^{-1}x)^2\,\mathrm{d}x$。

解.

Diff		Int
$(\sin^{-1}x)^2$	$+$	1
$\frac{2\sin^{-1}x}{\sqrt{1-x^2}}$	$-$	x

Diff		Int
$2\sin^{-1}x$	$+$	$\frac{x}{\sqrt{1-x^2}}$
$\frac{2}{\sqrt{1-x^2}}$	$-$	$-\sqrt{1-x^2}$

$$\begin{aligned}
&\int (\sin^{-1}x)^2\,\mathrm{d}x = (\sin^{-1}x)^2\cdot x - \int \frac{2\sin^{-1}x}{\sqrt{1-x^2}}\cdot x\,\mathrm{d}x \\
&= (\sin^{-1}x)^2\cdot x - \int 2\sin^{-1}x\cdot\frac{x}{\sqrt{1-x^2}}\,\mathrm{d}x \\
&= (\sin^{-1}x)^2\cdot x - \left(-2\sin^{-1}x\cdot\sqrt{1-x^2} + \int \frac{2}{\sqrt{1-x^2}}\cdot\sqrt{1-x^2}\,\mathrm{d}x\right) \\
&= (\sin^{-1}x)^2\cdot x + 2\sin^{-1}x\cdot\sqrt{1-x^2} - 2x
\end{aligned}$$

例. 求 $\int e^{ax}\cos bx\,\mathrm{d}x$，$a$、$b\neq 0$。

解.

Diff		Int
$\cos bx$	$+$	e^{ax}
$-b\sin bx$	$-$	$\frac{1}{a}e^{ax}$
$-b^2\cos bx$	$+$	$\frac{1}{a^2}e^{ax}$

$$\int e^{ax}\cos bx\,\mathrm{d}x = \frac{1}{a}e^{ax}\cos bx + \frac{b}{a^2}e^{ax}\sin bx - \frac{b^2}{a^2}\int e^{ax}\cos bx\,\mathrm{d}x$$
$$\Longrightarrow \int e^{ax}\cos bx\,\mathrm{d}x = \frac{e^{ax}(a\cos bx + b\sin bx)}{a^2+b^2}$$

例. 求 $\int e^{ax}\sin bx\,\mathrm{d}x$，$a$、$b \neq 0$。

解.

Diff		Int
$\sin bx$	$+$	e^{ax}
$b\cos bx$	$-$	$\frac{1}{a}e^{ax}$
$-b^2\sin bx$	$+$	$\frac{1}{a^2}e^{ax}$

$$\int e^{ax}\sin bx\,\mathrm{d}x = \frac{1}{a}e^{ax}\sin bx - \frac{b}{a^2}e^{ax}\cos bx - \frac{b^2}{a^2}\int e^{ax}\sin bx\,\mathrm{d}x$$
$$\Longrightarrow \int e^{ax}\sin bx\,\mathrm{d}x = \frac{e^{ax}(a\sin bx - b\cos bx)}{a^2+b^2}$$

遞迴式

例. 求 $I_n = \int \frac{1}{(x^2+a^2)^n}\,\mathrm{d}x$, $n \in \mathbb{N}$ 之遞迴式：令 $u = \frac{1}{(x^2+a^2)^n}$，則 $\mathrm{d}u = -2n\frac{x}{(x^2+a^2)^{n+1}}\,\mathrm{d}x$；$\mathrm{d}v = \mathrm{d}x$，則 $v = x$。故 $I_n = \int \frac{1}{(x^2+a^2)^n}\,\mathrm{d}x = \frac{x}{(x^2+a^2)^n} + 2n\int \frac{x^2}{(x^2+a^2)^{n+1}}\,\mathrm{d}x = \frac{x}{(x^2+a^2)^n} + 2n\int \frac{(x^2+a^2-a^2)}{(x^2+a^2)^{n+1}}\,\mathrm{d}x = \frac{x}{(x^2+a^2)^n} + 2nI_n - 2na^2 I_{n+1} \Longrightarrow 2na^2 I_{n+1} = \frac{x}{(x^2+a^2)^n} + (2n-1)I_n \Longrightarrow I_{n+1} = \frac{x}{2na^2(x^2+a^2)^n} + \frac{2n-1}{2na^2}I_n$。

使用例：$I_1 = \frac{1}{a}\tan^{-1}\frac{x}{a}$, $I_2 = \frac{1}{2a^2}I_1 + \frac{x}{2a^2(x^2+a^2)} = \frac{1}{2a^3}\tan^{-1}\frac{x}{a} + \frac{x}{2a^2(x^2+a^2)}$, $I_3 = \frac{3}{4a^2}I_2 + \frac{x}{4a^2(x^2+a^2)^2} = \frac{3}{4a^2}\left(\frac{1}{2a^3}\tan^{-1}\frac{x}{a} + \frac{x}{2a^2(x^2+a^2)}\right) + \frac{x}{4a^2(x^2+a^2)^2} = \frac{3}{8a^5}\tan^{-1}\frac{x}{a} + \frac{3x}{8a^4(x^2+a^2)} + \frac{x}{4a^2(x^2+a^2)^2}$。

例. 求 $J_n = \int \sin^n x\,\mathrm{d}x$，$n \in \mathbb{N}$，$n \geqslant 2$ 之遞迴式：令 $u = \sin^{n-1}x$，則 $\mathrm{d}u = (n-1)\sin^{n-2}x\cos x\,\mathrm{d}x$；$\mathrm{d}v = \sin x\,\mathrm{d}x$，則 $v = -\cos x$。故 $J_n = \int \sin^n x\,\mathrm{d}x = -\sin^{n-1}x\cdot\cos x + (n-1)\int \cos x\cdot\sin^{n-2}x\cos x\,\mathrm{d}x = -\sin^{n-1}x\cos x + (n-1)\int \sin^{n-2}x\cos^2 x\,\mathrm{d}x = -\sin^{n-1}x\cos x + (n-1)\int \sin^{n-2}x\cdot(1-\sin^2 x)\,\mathrm{d}x = -\sin^{n-1}x\cos x + (n-1)\int \sin^{n-2}x\,\mathrm{d}x - (n-1)\int \sin^n x\,\mathrm{d}x = -\sin^{n-1}x\cos x + (n-1)J_{n-2} + (1-n)J_n \Longrightarrow nJ_n = -\sin^{n-1}x\cos x + (n-1)J_{n-2} \Longrightarrow J_n = \frac{-\sin^{n-1}x\cos x}{n} + \frac{n-1}{n}J_{n-2}$。

使用例：$J_2 = \frac{-\sin x\cos x}{2} + \frac{1}{2}J_0 = \frac{-\sin x\cos x}{2} + \frac{x}{2}$，$J_3 = \frac{-\sin^2 x\cos x}{3} + \frac{2}{3}J_1 = \frac{-\sin^2 x\cos x}{3} - \frac{2\cos x}{3}$，$J_4 = \frac{-\sin^3 x\cos x}{4} + \frac{3}{4}J_2 = \frac{-\sin^3 x\cos x}{4} + \frac{3\sin x\cos x}{8} - \frac{3x}{8}$。

習題. 以部份積分法求下列不定積分。注意：可能會因為常數項而跟此處答案不同。

1. $\int x^3\ln x\,\mathrm{d}x = \frac{x^4}{4}\ln x - \frac{x^4}{16} + c$
2. $\int x(\ln x)^3\,\mathrm{d}x = \frac{x^2}{2}\left((\ln x)^3 - \frac{3(\ln x)^2}{2} + \frac{3\ln x}{2} - \frac{3}{4}\right) + c$
3. $\int x^2\tan^{-1}x\,\mathrm{d}x = \frac{x^3}{3}\tan^{-1}x - \frac{x^2}{6} + \frac{\ln(1+x^2)}{6} + c$
4. $\int x^5 e^{-x^2}\,\mathrm{d}x = -\frac{1}{2}e^{-x^2}(x^4+2x^2+2) + c$

5. $\displaystyle\int xe^{\sqrt{x}}\,\mathrm{d}x = 2\,e^{\sqrt{x}}\left(x\sqrt{x}-3x+6\sqrt{x}-6\right)+c$

6. $\displaystyle\int \frac{xe^x}{(x+1)^2}\,\mathrm{d}x = \frac{e^x}{x+1}+c$

7. $\displaystyle\int \sin(\ln x)\,\mathrm{d}x = \frac{x\left(\sin(\ln x)-\cos(\ln x)\right)}{2}+c$

8. $\displaystyle\int \frac{\ln x}{\sqrt{1+x}}\,\mathrm{d}x = \ln x\cdot 2\sqrt{1+x}-4\sqrt{1+x}-2\ln(\sqrt{1+x}-1)+2\ln(\sqrt{1+x}+1)+c$

解.

1. 令 $u=\ln x$，則 $\mathrm{d}u=\dfrac{1}{x}\,\mathrm{d}x$；$\mathrm{d}v=x^3\,\mathrm{d}x$，則 $v=\dfrac{x^4}{4}$。故 $\displaystyle\int x^3\ln x\,\mathrm{d}x=\ln x\cdot\frac{x^4}{4}-\int\frac{x^4}{4}\cdot\frac{1}{x}\,\mathrm{d}x=\ln x\cdot\frac{x^4}{4}-\int\frac{x^3}{4}\,\mathrm{d}x=\ln x\cdot\frac{x^4}{4}-\frac{x^4}{16}+c$。

2. 令 $u=(\ln x)^3$，則 $\mathrm{d}u=3(\ln x)^2\cdot\dfrac{1}{x}\,\mathrm{d}x$；$\mathrm{d}v=x\,\mathrm{d}x$，則 $v=\dfrac{x^2}{2}$。故 $\displaystyle\int x(\ln x)^3\,\mathrm{d}x=(\ln x)^3\cdot\frac{x^2}{2}-\int\frac{x^2}{2}\cdot 3(\ln x)^2\cdot\frac{1}{x}\,\mathrm{d}x=(\ln x)^3\cdot\frac{x^2}{2}-\frac{3}{2}\int x(\ln x)^2\,\mathrm{d}x$。令 $u=(\ln x)^2$，則 $\mathrm{d}u=2\,\ln x\cdot\dfrac{1}{x}\,\mathrm{d}x$；$\mathrm{d}v=x\,\mathrm{d}x$，則 $v=\dfrac{x^2}{2}$。故 $\displaystyle\int x(\ln x)^2\,\mathrm{d}x=(\ln x)^2\cdot\frac{x^2}{2}-\int\frac{x^2}{2}\cdot 2\ln x\cdot\frac{1}{x}\,\mathrm{d}x=(\ln x)^2\cdot\frac{x^2}{2}-\int x\ln x\,\mathrm{d}x$。令 $u=\ln x$，則 $\mathrm{d}u=\dfrac{1}{x}\,\mathrm{d}x$；$\mathrm{d}v=x\,\mathrm{d}x$，則 $v=\dfrac{x^2}{2}$。故 $\displaystyle\int x\ln x\,\mathrm{d}x=\ln x\cdot\frac{x^2}{2}-\int\frac{x^2}{2}\cdot\frac{1}{x}\,\mathrm{d}x=\ln x\cdot\frac{x^2}{2}-\frac{1}{2}\int x\,\mathrm{d}x=\ln x\cdot\frac{x^2}{2}-\frac{x^2}{4}$。以上，$\displaystyle\int x(\ln x)^3\,\mathrm{d}x=(\ln x)^3\cdot\frac{x^2}{2}-\frac{3}{2}\Big((\ln x)^2\cdot\frac{x^2}{2}-\big(\ln x\cdot\frac{x^2}{2}-\frac{x^2}{4}\big)\Big)=\frac{x^2}{2}\big((\ln x)^3-\frac{3(\ln x)^2}{2}+\frac{3\ln x}{2}-\frac{3}{4}\big)+c$

3. 令 $u=\tan^{-1}x$，則 $\mathrm{d}u=\dfrac{1}{1+x^2}\,\mathrm{d}x$；$\mathrm{d}v=x^2\,\mathrm{d}x$，則 $v=\dfrac{x^3}{3}$。故 $\displaystyle\int x^2\tan^{-1}x\,\mathrm{d}x=\tan^{-1}x\cdot\frac{x^3}{3}-\int\frac{x^3}{3}\cdot\frac{1}{x^2+1}\,\mathrm{d}x$。又 $\displaystyle\int\frac{x^3}{3}\cdot\frac{1}{x^2+1}\,\mathrm{d}x=\frac{1}{3}\int\frac{(x^3+x)-x}{x^2+1}\,\mathrm{d}x=\frac{1}{3}\int\frac{x(x^2+1)-x}{x^2+1}\,\mathrm{d}x=\frac{1}{3}\int\big(x-\frac{x}{x^2+1}\big)\,\mathrm{d}x$。令 $w=x^2+1$，則 $\mathrm{d}w=2x\,\mathrm{d}x\implies x\,\mathrm{d}x=\dfrac{1}{2}\,\mathrm{d}w$，故 $\displaystyle\int\frac{x}{x^2+1}\,\mathrm{d}x=\int\frac{1}{x^2+1}\cdot x\,\mathrm{d}x=\int\frac{1}{w}\cdot\frac{1}{2}\,\mathrm{d}w=\frac{1}{2}\ln w+c=\frac{1}{2}\ln(x^2+1)+c$。以上，$\displaystyle\int x^2\tan^{-1}x\,\mathrm{d}x=\tan^{-1}x\cdot\frac{x^3}{3}-\int\frac{x^3}{3}\cdot\frac{1}{x^2+1}\,\mathrm{d}x=\tan^{-1}x\cdot\frac{x^3}{3}-\frac{x^2}{6}+\frac{\ln(x^2+1)}{6}+c$。

4. 令 $w=x^2$，則 $\mathrm{d}w=2x\,\mathrm{d}x\implies x\,\mathrm{d}x=\dfrac{1}{2}\,\mathrm{d}w$，故 $\displaystyle\int x^5e^{-x^2}\,\mathrm{d}x=\int e^{-x^2}\cdot(x^2)^2\cdot x\,\mathrm{d}x=\int e^{-w}\cdot w^2\cdot\frac{1}{2}\,\mathrm{d}w=\frac{1}{2}\int w^2\,e^{-w}\,\mathrm{d}w=-\frac{1}{2}\,e^{-w}(w^2+2w+2)=-\frac{1}{2}\,e^{-x^2}(x^4+2x^2+2)+c$。

Diff		Int
w^2	$+$	e^{-w}
$2w$	$-$	$-e^{-w}$
2	$+$	e^{-w}
0		$-e^{-w}$

$$\int w^2e^{-w}\,\mathrm{d}w=-w^2\,e^{-w}-2w\,e^{-w}-2\,e^{-w}=-e^{-w}(w^2+2w+2)$$

5. 令 $w=\sqrt{x}$，則 $w^2=x$，$\mathrm{d}x=2w\,\mathrm{d}w$，故 $\displaystyle\int xe^{\sqrt{x}}\,\mathrm{d}x=\int w^2e^w\cdot 2w\,\mathrm{d}w=2\int w^3e^w\,\mathrm{d}w=2\,e^w(w^3-3w^2+6w-6)=2\,e^{\sqrt{x}}\left(x\sqrt{x}-3x+6\sqrt{x}-6\right)+c$

Diff		Int
w^3	$+$	e^w
$3w^2$	$-$	e^w
$6w$	$+$	e^w
6	$-$	e^w
0		e^w

$$\int w^3\,e^w\,\mathrm{d}w = w^3\,e^w - 3w^2\,e^w + 6w\,e^w - 6\,e^w = e^w(w^3 - 3w^2 + 6w - 6)$$

6. $\displaystyle\int \frac{xe^x}{(x+1)^2}\,\mathrm{d}x = \int \frac{(x+1-1)e^x}{(x+1)^2}\,\mathrm{d}x = \int \frac{e^x}{x+1}\,\mathrm{d}x - \int \frac{e^x}{(x+1)^2}\,\mathrm{d}x$。令 $u = e^x$，則 $\mathrm{d}u = e^x\,\mathrm{d}x$；$\mathrm{d}v = \dfrac{1}{(x+1)^2}\,\mathrm{d}x$，則 $v = \dfrac{-1}{x+1}$。故 $\displaystyle\int \frac{e^x}{(x+1)^2}\,\mathrm{d}x = e^x \cdot \frac{-1}{x+1} + \int \frac{1}{x+1} \cdot e^x\,\mathrm{d}x = \frac{-e^x}{x+1} + \int \frac{e^x}{x+1}\,\mathrm{d}x$；原式 $\displaystyle\int \frac{xe^x}{(x+1)^2}\,\mathrm{d}x = \int \frac{e^x}{x+1}\,\mathrm{d}x - \int \frac{e^x}{(x+1)^2}\,\mathrm{d}x = \int \frac{e^x}{x+1}\,\mathrm{d}x - \left(\frac{-e^x}{x+1} + \int \frac{e^x}{x+1}\,\mathrm{d}x\right) = \frac{e^x}{x+1}$。

7. 令 $w = \ln x$，則 $e^w = x$，$\mathrm{d}x = e^w\,\mathrm{d}w$，故 $\displaystyle\int \sin(\ln x)\,\mathrm{d}x = \int \sin w \cdot e^w\,\mathrm{d}w = \int e^w \sin w\,\mathrm{d}w = \frac{e^w(\sin w - \cos w)}{2} = \frac{x\,(\sin(\ln x) - \cos(\ln x))}{2} + c$

Diff		Int
$\sin w$	$+$	e^w
$\cos w$	$-$	e^w
$-\sin w$	$+$	e^w

$$\int e^w \sin w\,\mathrm{d}w = e^w \sin w - e^w \cos w - \int e^w \sin w\,\mathrm{d}w$$

$$\Longrightarrow \int e^w \sin w\,\mathrm{d}w = \frac{e^w(\sin w - \cos w)}{2}$$

8. 令 $u = \ln x$，則 $\mathrm{d}u = \dfrac{1}{x}\,\mathrm{d}x$；$\mathrm{d}v = \dfrac{1}{\sqrt{1+x}}\,\mathrm{d}x$，則 $v = 2\sqrt{1+x}$。故 $\displaystyle\int \frac{\ln x}{\sqrt{1+x}}\,\mathrm{d}x = \ln x \cdot 2\sqrt{1+x} - 2\int \frac{\sqrt{1+x}}{x}\,\mathrm{d}x$。令 $w = \sqrt{1+x}$，則 $w^2 = 1 + x \implies x = w^2 - 1$，$2w\,\mathrm{d}w = \mathrm{d}x$，故 $\displaystyle\int \frac{\sqrt{1+x}}{x}\,\mathrm{d}x = \int \frac{w}{w^2-1} \cdot 2w\,\mathrm{d}w = 2\int \frac{w^2}{w^2-1}\,\mathrm{d}w = 2\int \frac{(w^2-1)+1}{w^2-1}\,\mathrm{d}w = 2w + 2\int \frac{1}{w^2-1}\,\mathrm{d}w = 2\sqrt{1+x} + 2\int \frac{1}{w^2-1}\,\mathrm{d}w$。由部份分式 $\dfrac{1}{w^2-1} = \dfrac{1}{2}\left(\dfrac{1}{w-1} - \dfrac{1}{w+1}\right)$，$\displaystyle\int \frac{1}{w^2-1}\,\mathrm{d}w = \frac{1}{2}\int \left(\frac{1}{w-1} - \frac{1}{w+1}\right)\mathrm{d}w = \frac{1}{2}\left(\ln(w-1) - \ln(w+1)\right) = \frac{1}{2}\left(\ln(\sqrt{1+x}-1) - \ln(\sqrt{1+x}+1)\right)$。以上，$\displaystyle\int \frac{\ln x}{\sqrt{1+x}}\,\mathrm{d}x = \ln x \cdot 2\sqrt{1+x} - 2\left(2\sqrt{1+x} + 2\left(\frac{1}{2}\left(\ln(\sqrt{1+x}-1) - \ln(\sqrt{1+x}+1)\right)\right)\right) = \ln x \cdot 2\sqrt{1+x} - 4\sqrt{1+x} - 2\ln(\sqrt{1+x}-1) + 2\ln(\sqrt{1+x}+1) + c$。

4.2 定積分

定積分 $\approx$ 面積：x 軸上方為正，下方為負。

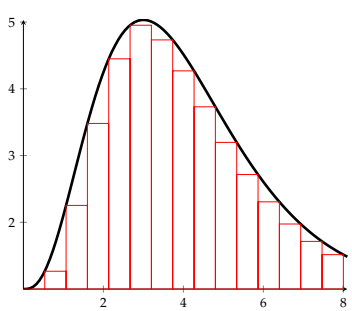
定義. 給定 $f : [a,\, b] \to \mathbb{R}$。

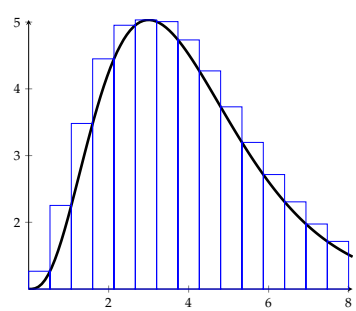
- $[a,\, b]$ 分割 $\mathbb{P} : a = x_0 < x_1 < x_2 < \cdots < x_n = b$
- $\Delta x_k = x_k - x_{k-1}$，$k = 1,\, 2,\, \ldots,\, n$；$\|\mathbb{P}\| = \max\{\, |\Delta x_k| \mid 1 \leqslant k \leqslant n\}$
- 樣本點 ξ_k：$x_{k-1} \leqslant \xi_k \leqslant x_k$，$k = 1,\, 2,\, \ldots,\, n$
- $u_k = \sup\{\, f(x) \mid x_{k-1} \leqslant x \leqslant x_k\}$，$l_k = \inf\{\, f(x) \mid x_{k-1} \leqslant x \leqslant x_k\}$，$k = 1,\, 2,\, \ldots,\, n$
- $R(f, \mathbb{P}) = \displaystyle\sum_{k=1}^{n} f(\xi_k)\, \Delta x_k$，$U(f, \mathbb{P}) = \displaystyle\sum_{k=1}^{n} u_k\, \Delta x_k$，$L(f, \mathbb{P}) = \displaystyle\sum_{k=1}^{n} l_k\, \Delta x_k$；
  顯然 $L(f, \mathbb{P}) \leqslant R(f, \mathbb{P}) \leqslant U(f, \mathbb{P})$。
- 求 $\displaystyle\lim_{\|\mathbb{P}\| \to 0} R(f, \mathbb{P})$。若對不同分割與樣本點選取此極限均存在且相等，稱 f 在 $[a, b]$ 可積（分）；$f(x)$ 在 $[a,\, b]$ 的定積分 $\displaystyle\int_a^b f(x)\, \mathrm{d}x \equiv \lim_{\|\mathbb{P}\| \to 0} R(f, \mathbb{P})$

註.

- 在 $\displaystyle\int_a^b f(x)\, \mathrm{d}x$ 中，a 為*積分下限* (lower limit of integration)，b 為*積分上限* (upper limit of integration)，$f(x)$ 為*被積分式* (integrand)，x 為*積分變數* (variable of integration)。
- $\displaystyle\int_a^b f(x)\, \mathrm{d}x = \int_a^b f(t)\, \mathrm{d}t = \int_a^b f(u)\, \mathrm{d}u$

結論. 若 f 在 $[a,\, b]$ 連續，則 f 在 $[a,\, b]$ 可積。





性質. 令 f、g 在包含 a、b、c 之區間為可積，α、$\beta \in \mathbb{R}$。則

1. $\displaystyle\int_a^a f(x)\, \mathrm{d}x = 0$
2. $\displaystyle\int_a^b f(x)\, \mathrm{d}x = -\int_b^a f(x)\, \mathrm{d}x$
3. $\displaystyle\int_a^b \big(\alpha\, f(x) + \beta\, g(x)\big)\, \mathrm{d}x = \alpha \int_a^b f(x)\, \mathrm{d}x + \beta \int_a^b g(x)\, \mathrm{d}x$
4. $\displaystyle\int_a^b f(x)\, \mathrm{d}x + \int_b^c f(x)\, \mathrm{d}x = \int_a^c f(x)\, \mathrm{d}x$
5. $\displaystyle\int_a^b f(x)\, \mathrm{d}x \leqslant \int_a^b g(x)\, \mathrm{d}x$，若 $f(x) \leqslant g(x)\ \forall\, x \in [a, b],\ a \leqslant b$

6. $\left|\int_a^b f(x)\,\mathrm{d}x\right| \leqslant \int_a^b |\,f(x)\,|\,\mathrm{d}x, \quad a \leqslant b$

7. $f(x)$ 為奇函數：$\int_{-a}^{a} f(x)\,\mathrm{d}x = 0$

8. $f(x)$ 為偶函數：$\int_{-a}^{a} f(x)\,\mathrm{d}x = 2\int_0^a f(x)\,\mathrm{d}x$

例.

1. $\int_{-\sqrt{2}}^{\sqrt{2}} \sqrt{2-x^2}\,\mathrm{d}x = \pi$（半徑 $\sqrt{2}$ 之半圓面積）
2. $\int_{-1}^{2} \operatorname{sgn}(x)\,\mathrm{d}x = 2\cdot 1 - 1\cdot 1 = 1$
3. $\int_{-\pi}^{\pi} \sin(x^7-5x^3)\,\mathrm{d}x = 0$（$\sin(x^7-5x^3)$ 為奇函數）
4. $\int_{-6}^{6} e^{-x^4}\sin(\sin x)\,\mathrm{d}x = 0$（$e^{-x^4}\sin(\sin x)$ 為奇函數）
5. $\int_{-4}^{4} \left(e^x - e^{-x}\right)\mathrm{d}x = 0$（$e^x - e^{-x}$ 為奇函數）
6. $\int_{-2024}^{2024} \left(e^{9x^5-2x^7} - e^{-9x^5+2x^7}\right)\mathrm{d}x = 0$（$e^{9x^5-2x^7} - e^{-9x^5+2x^7}$ 為奇函數）
7. $\int_{-a}^{a} |x|\,\mathrm{d}x = 2\int_0^a |x|\,\mathrm{d}x = a^2$（$|x|$ 為偶函數；兩 $a\times a$ 等腰直角三角形面積）
8. 定義 $\int_1^x \frac{1}{\tau}\,\mathrm{d}\tau \equiv \ln x$，則 $\int_{\frac{1}{4}}^{3} \frac{1}{x}\,\mathrm{d}x = \int_{\frac{1}{4}}^{1} \frac{1}{x}\,\mathrm{d}x + \int_1^3 \frac{1}{x}\,\mathrm{d}x = -\int_1^{\frac{1}{4}} \frac{1}{x}\,\mathrm{d}x + \int_1^3 \frac{1}{x}\,\mathrm{d}x = \ln 12$
9. $\int_0^2 \sqrt{4-x^2}\cdot \operatorname{sgn}(1-x)\,\mathrm{d}x = \sqrt{3} - \frac{\pi}{3}$
10. $\forall\, n \in \mathbb{N},\ \int_n^{n+1} \lfloor x \rfloor\,\mathrm{d}x = n$

以極限定義求定積分

結論. $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$。

例. 求 $\int_0^1 x^2\,\mathrm{d}x$。

解. 建立 $[0,\,1]$ 分割 $\mathbb{P}:\left\{0, \frac{1}{n}, \frac{2}{n}, \ldots, \frac{n}{n}\right\}$，則 $\Delta x_k = \frac{1}{n}\ \forall\, k = 1, 2, \ldots, n$ 且 $\|P\| \to 0$ 當 $n \to \infty$。$f(x) = x^2$ 且 f 在 $[0,\,1]$ 嚴格遞增，$U(f,\mathbb{P}) = \sum_{k=1}^{n} u_k\,\Delta x_k = \sum_{k=1}^{n} \frac{k^2}{n^2}\cdot\frac{1}{n} = \frac{1}{n^3}\sum_{k=1}^{n} k^2$，$L(f,\mathbb{P}) = \sum_{k=1}^{n} l_k\,\Delta x_k = \sum_{k=1}^{n} \frac{(k-1)^2}{n^2}\cdot\frac{1}{n} = \frac{1}{n^3}\sum_{k=1}^{n-1} k^2$，$L(f,\mathbb{P}) \leqslant R(f,\mathbb{P}) \leqslant U(f,\mathbb{P})$。由 $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$，$\lim_{n\to\infty} U(f,\mathbb{P}) = \lim_{n\to\infty} \frac{1}{n^3}\sum_{k=1}^{n} k^2 =$

$\lim\limits_{n\to\infty}\dfrac{n(n+1)(2n+1)}{6n^3}=\dfrac{1}{3}$，$\lim\limits_{n\to\infty}L(f,\mathbb{P})=\lim\limits_{n\to\infty}\dfrac{1}{n^3}\sum\limits_{k=1}^{n-1}k^2=\lim\limits_{n\to\infty}\dfrac{(n-1)n(2n-1)}{6n^3}=\dfrac{1}{3}$，故由夾擊定理 $\lim\limits_{n\to\infty}R(f,\mathbb{P})=\dfrac{1}{3}$。

結論. 若 $r\in\mathbb{R}$，$\displaystyle\sum_{k=0}^{n}r^k=\frac{r^{n+1}-1}{r-1}$。

例. 求 $\displaystyle\int_0^1 e^x\,\mathrm{d}x$。

解. 建立 $[0,\,1]$ 分割 $\mathbb{P}:\left\{0,\dfrac{1}{n},\dfrac{2}{n},\ldots,\dfrac{n}{n}\right\}$，則 $\Delta x_k=\dfrac{1}{n}\ \forall\,k=1,2,\ldots,n$ 且 $\|P\|\to 0$ 當 $n\to\infty$。$f(x)=e^x$ 且 f 在 $[0,\,1]$ 嚴格遞增，$U(f,\mathbb{P})=\displaystyle\sum_{k=1}^{n}u_k\,\Delta x_k=\sum_{k=1}^{n}e^{\frac{k}{n}}\cdot\frac{1}{n}=\frac{1}{n}\frac{e^{\frac{1}{n}}\left(e^{\frac{n}{n}}-1\right)}{e^{\frac{1}{n}}-1}=\frac{e-1}{n}\frac{e^{\frac{1}{n}}}{e^{\frac{1}{n}}-1}$，$L(f,\mathbb{P})=\displaystyle\sum_{k=1}^{n}l_k\,\Delta x_k=\sum_{k=1}^{n}e^{\frac{k-1}{n}}\cdot\frac{1}{n}=\frac{1}{n}\frac{e^0\left(e^{\frac{n}{n}}-1\right)}{e^{\frac{1}{n}}-1}=\frac{e-1}{n}\frac{1}{e^{\frac{1}{n}}-1}$，$L(f,\mathbb{P})\leqslant R(f,\mathbb{P})\leqslant U(f,\mathbb{P})$。$\lim\limits_{n\to\infty}U(f,\mathbb{P})=\lim\limits_{n\to\infty}\dfrac{e-1}{n}\dfrac{e^{\frac{1}{n}}}{e^{\frac{1}{n}}-1}=(e-1)\lim\limits_{x\to 0+}\dfrac{xe^x}{e^x-1}=(e-1)\lim\limits_{x\to 0+}\dfrac{e^x+xe^x}{e^x}=e-1$，$\lim\limits_{n\to\infty}L(f,\mathbb{P})=\lim\limits_{n\to\infty}\dfrac{e-1}{n}\dfrac{1}{e^{\frac{1}{n}}-1}=(e-1)\lim\limits_{x\to 0+}\dfrac{x}{e^x-1}=(e-1)\lim\limits_{x\to 0+}\dfrac{1}{e^x}=e-1$，故由夾擊定理 $\lim\limits_{n\to\infty}R(f,\mathbb{P})=e-1$。

結論. $\displaystyle\sum_{k=1}^{n}\cos kx=\frac{1}{2}\left(\frac{\sin\left(n+\frac{1}{2}\right)x}{\sin\frac{x}{2}}-1\right)$

證. 考慮 $\displaystyle\sum_{k=1}^{n}e^{ikx}=\frac{e^{ix}(1-e^{inx})}{1-e^{ix}}$,則 $\displaystyle\sum_{k=1}^{n}\cos kx=\Re\left\{\sum_{k=1}^{n}e^{ikx}\right\}=\Re\left\{\frac{e^{ix}(1-e^{inx})}{1-e^{ix}}\right\}=\Re\left\{\frac{e^{ix}\,e^{\frac{inx}{2}}\left(e^{-\frac{inx}{2}}-e^{\frac{inx}{2}}\right)}{e^{\frac{ix}{2}}\left(e^{-\frac{ix}{2}}-e^{\frac{ix}{2}}\right)}\right\}=\frac{\cos\frac{(n+1)x}{2}\sin\frac{nx}{2}}{\sin\frac{x}{2}}=\frac{\sin\left(\frac{nx}{2}+\frac{(n+1)x}{2}\right)+\sin\left(\frac{nx}{2}+\frac{(n+1)x}{2}\right)}{2\,\sin\frac{x}{2}}=\frac{\sin\left(n+\frac{1}{2}\right)x-\sin\frac{x}{2}}{2\,\sin\frac{x}{2}}=\frac{1}{2}\left(\frac{\sin\left(n+\frac{1}{2}\right)x}{\sin\frac{x}{2}}-1\right)$。

例. 求 $\displaystyle\int_0^{\frac{\pi}{2}}\cos x\,\mathrm{d}x$。

解. 建立 $\left[0,\,\dfrac{\pi}{2}\right]$ 分割 $\mathbb{P}:\left\{0,\dfrac{\pi}{2n},\dfrac{2\pi}{2n},\ldots,\dfrac{n\pi}{2n}\right\}$，則 $\Delta x_k=\dfrac{\pi}{2n}\ \forall\,k=1,2,\ldots,n$ 且 $\|P\|\to 0$ 當 $n\to\infty$，則積分為 $\lim\limits_{n\to\infty}R(f,\mathbb{P})=\displaystyle\sum_{k=1}^{n}f(\xi_k)\,\Delta x_k=\lim_{n\to\infty}\sum_{k=1}^{n}\cos\frac{k\pi}{2n}\cdot\frac{\pi}{2n}=\lim_{n\to\infty}\sum_{k=1}^{n}\cos\left(k\left(\frac{\pi}{2n}\right)\right)\cdot\frac{\pi}{2n}=\lim_{n\to\infty}\frac{1}{2}\left(\frac{\sin\left(\left(n+\frac{1}{2}\right)\cdot\frac{\pi}{2n}\right)}{\sin\left(\frac{1}{2}\cdot\frac{\pi}{2n}\right)}-1\right)\cdot\frac{\pi}{2n}=\lim_{n\to\infty}\frac{\sin\left(\left(n+\frac{1}{2}\right)\frac{\pi}{2n}\right)\frac{1}{2}\cdot\frac{\pi}{2n}}{\sin\left(\frac{1}{2}\cdot\frac{\pi}{2n}\right)}=\lim_{n\to\infty}\sin\left(\frac{1}{2}+\frac{1}{4n}\right)\pi\cdot\lim_{n\to\infty}\frac{\frac{1}{2}\cdot\frac{\pi}{2n}}{\sin\left(\frac{1}{2}\cdot\frac{\pi}{2n}\right)}=\sin\frac{\pi}{2}\cdot 1=1.$

微積分基本定理

定理 (微積分基本定理（Fundamental Theorem of Calculus，FTC）).

1. 若 f 在 $[a,\,b]$ 連續，令 $F(x)=\displaystyle\int_a^x f(\tau)\,\mathrm{d}\tau$ 且 $a\leqslant x\leqslant b$，則 $F'(x)=f(x)\ \forall\,x\in[a,\,b]$。
2. 若 $G'(x)=f(x)\ \forall\,x\in[a,\,b]$，則 $\displaystyle\int_a^b f(x)\,\mathrm{d}x=G(b)-G(a)\equiv G(x)\Big|_a^b$。

註. 由 FTC，$\displaystyle\int_a^b f(x)\,\mathrm{d}x$ 可由 f 的反導函數（不定積分）得出，不需繁複極限計算！

證.

1. WLOG 令 $h>0$。則 $\left|\dfrac{F(x+h)-F(x)}{h}-f(x)\right| = \left|\dfrac{1}{h}\displaystyle\int_x^{x+h}(f(\tau)-f(x))\,\mathrm{d}\tau\right| \leqslant \dfrac{1}{h}\displaystyle\int_x^{x+h}\big|f(\tau)-f(x)\big|\,\mathrm{d}\tau \leqslant \max_{x\leqslant\tau\leqslant x+h}\big|f(\tau)-f(x)\big| \to 0$，當 $h\to 0$ 且 f 為連續。
2. $(G(x)-F(x))' = f(x)-f(x) = 0\ \forall x\in[a,b]$，故 $G(x)-F(x)$ 在 $[a,b]$ 為常數 $\implies G(a)-F(a) = G(b)-F(b) \implies G(b)-G(a) = F(b)-F(a) = \displaystyle\int_a^b f(x)\,\mathrm{d}x$。

例 (以 FTC 求定積分).

1. x^2 之反導函數為 $\dfrac{x^3}{3}$，故 $\displaystyle\int_0^1 x^2\,\mathrm{d}x = \frac{1^3}{3}-\frac{0^3}{3} = \frac{1}{3}$。
2. e^x 之反導函數為 e^x，故 $\displaystyle\int_0^1 e^x\,\mathrm{d}x = e^1-e^0 = e-1$。
3. $\cos x$ 之反導函數為 $\sin x$，故 $\displaystyle\int_0^{\frac{\pi}{2}}\cos x\,\mathrm{d}x = \sin\frac{\pi}{2}-\sin 0 = 1$。

結論 (定積分變數變換).

- 求反導函數後代入：$\displaystyle\int_a^b f'(g(x))\,g'(x)\,\mathrm{d}x = f(g(x))\Big|_{x=a}^{x=b} = f(g(b))-f(g(a))$
- 變數變換並改變積分範圍：$\displaystyle\int_a^b f'(g(x))\,g'(x)\,\mathrm{d}x = \int_a^b f'(g(x))\,\mathrm{d}g(x) = \int_{g(a)}^{g(b)} f'(u)\,\mathrm{d}u = f(u)\Big|_{u=g(a)}^{u=g(b)} = f(g(b))-f(g(a))$

例. 求 $\displaystyle\int_0^1 x^3(1+x^4)^3\,\mathrm{d}x$。

解.

- 求反導函數後代入：令 $u=1+x^4$，則 $\mathrm{d}u = 4x^3\,\mathrm{d}x \implies x^3\,\mathrm{d}x = \dfrac{\mathrm{d}u}{4}$，故 $\displaystyle\int x^3(1+x^4)^3\,\mathrm{d}x = \int(1+x^4)^3\,x^3\,\mathrm{d}x = \int u^3\,\frac{\mathrm{d}u}{4} = \frac{u^4}{16}+c = \frac{(1+x^4)^4}{16}+c$。故 $\displaystyle\int_0^1 x^3(1+x^4)^3\,\mathrm{d}x = \frac{(1+x^4)^4}{16}\Big|_{x=0}^{x=1} = \frac{(1+1^4)^4-(1+0^4)^4}{16} = \frac{15}{16}$。
- 變數變換並改變積分範圍：令 $u=1+x^4$，則 $\mathrm{d}u = 4x^3\,\mathrm{d}x \implies x^3\,\mathrm{d}x = \dfrac{\mathrm{d}u}{4}$。積分範圍 x 由 0 至 1，則變數變換後 u 由 $1+0^4=1$ 至 $1+1^4=2$，故 $\displaystyle\int_0^1 x^3(1+x^4)^3\,\mathrm{d}x = \int_0^1(1+x^4)^3\,x^3\,\mathrm{d}x = \int_1^2 u^3\,\frac{\mathrm{d}u}{4} = \frac{1}{4}\int_1^2 u^3\,\mathrm{d}u = \frac{1}{16}u^4\Big|_{u=1}^{u=2} = \frac{2^4-1^4}{16} = \frac{15}{16}$。

例. 求 $\displaystyle\int_0^{\sqrt{3}}\frac{4x}{\sqrt{1+x^2}}\,\mathrm{d}x$。

解.

- 求反導函數後代入：令 $u=1+x^2$，則 $\mathrm{d}u = 2x\,\mathrm{d}x \implies 4x\,\mathrm{d}x = 2\,\mathrm{d}u$，故 $\displaystyle\int\frac{4x}{\sqrt{1+x^2}}\,\mathrm{d}x = \int\frac{2}{\sqrt{u}}\,\mathrm{d}u = 4\sqrt{u}+c = 4\sqrt{1+x^2}+c$。故 $\displaystyle\int_0^{\sqrt{3}}\frac{4x}{\sqrt{1+x^2}}\,\mathrm{d}x = 4\sqrt{1+x^2}\Big|_{x=0}^{x=\sqrt{3}} = 4\sqrt{1+3}-4\sqrt{1} = 4$。

- 變數變換並改變積分範圍：令 $u = 1 + x^2$，則 $\mathrm{d}u = 2x\,\mathrm{d}x \implies 4x\,\mathrm{d}x = 2\,\mathrm{d}u$。積分範圍 x 由 0 至 $\sqrt{3}$，則變數變換後 u 由 $1 + 0^2 = 1$ 至 $1 + (\sqrt{3})^2 = 4$，故 $\displaystyle\int_0^{\sqrt{3}} \frac{4x}{\sqrt{1+x^2}}\,\mathrm{d}x = \int_1^4 \frac{2}{\sqrt{u}}\,\mathrm{d}u = 4\sqrt{u}\Big|_{u=1}^{u=4} = 4\,(\sqrt{4} - \sqrt{1}) = 4$。

例. 求 $\displaystyle\int_0^{\pi} 3\cos^2 x \sin x\,\mathrm{d}x$。

解.

- 求反導函數後代入：令 $u = \cos x$，則 $\mathrm{d}u = -\sin x\,\mathrm{d}x \implies \sin x\,\mathrm{d}x = -\mathrm{d}u$，故 $\displaystyle\int 3\cos^2 x \sin x\,\mathrm{d}x = -3\int u^2\,\mathrm{d}u = -u^3 + c = -\cos^3 x + c$。故 $\displaystyle\int_0^{\pi} 3\cos^2 x \sin x\,\mathrm{d}x = -\cos^3 x\Big|_{x=0}^{x=\pi} = -(\cos^3\pi - \cos^3 0) = -((-1)^3 - 1^3) = 2$。
- 變數變換並改變積分範圍：令 $u = \cos x$，則 $\mathrm{d}u = -\sin x\,\mathrm{d}x \implies \sin x\,\mathrm{d}x = -\mathrm{d}u$。積分範圍 x 由 0 至 π，則變數變換後 u 由 $\cos 0 = 1$ 至 $\cos\pi = -1$，故 $\displaystyle\int_0^{\pi} 3\cos^2 x \sin x\,\mathrm{d}x = -\int_1^{-1} 3u^2\,\mathrm{d}u = -u^3\Big|_{u=1}^{u=-1} = -((-1)^3 - 1^3) = 2$。

性質. 令 $\displaystyle F(x) = \int_{v(x)}^{u(x)} f(\tau)\,\mathrm{d}\tau$，則 $F'(x) = f(u(x)) \cdot u'(x) - f(v(x)) \cdot v'(x)$。

證. 令 $a \in \mathbb{R}$, $\displaystyle F(x) = \int_{v(x)}^{u(x)} f(\tau)\,\mathrm{d}\tau = \int_a^{u(x)} f(\tau)\,\mathrm{d}\tau - \int_a^{v(x)} f(\tau)\,\mathrm{d}\tau$。令 $\displaystyle G(x) \equiv \int_a^x f(\tau)\,\mathrm{d}\tau$, 則 $G'(x) = f(x)$, $\displaystyle F(x) = \int_a^{u(x)} f(\tau)\,\mathrm{d}\tau - \int_a^{v(x)} f(\tau)\,\mathrm{d}\tau = G(u(x)) - G(v(x))$；故 $F'(x) = (G(u(x)) - G(v(x)))' = G'(u(x)) \cdot u'(x) - G'(v(x)) \cdot v'(x) = f(u(x)) \cdot u'(x) - f(v(x)) \cdot v'(x)$。

註.

例.

1. $\displaystyle F(x) = \int_1^x \frac{1}{1+\tau^4}\,\mathrm{d}\tau \implies F'(x) = \frac{1}{1+x^4}$
2. $\displaystyle F(x) = \int_2^{\sqrt{x}} \sin\tau\,\mathrm{d}\tau \implies F'(x) = \sin\sqrt{x} \cdot \frac{1}{2\sqrt{x}}$
3. $\displaystyle F(x) = \int_x^{2x} \tau^3\,\mathrm{d}\tau \implies F'(x) = (2x)^3 \cdot 2 - x^3 \cdot 1 = 15x^3$
4. $\displaystyle F(x) = \int_{\sin x}^{\tan^{-1} x} e^{\tau^2}\,\mathrm{d}\tau \implies F'(x) = e^{(\tan^{-1} x)^2} \cdot \frac{1}{1+x^2} - e^{\sin^2 x} \cdot \cos x$

例. 若 $\displaystyle g(x) = \int_0^{\cos x} (1 + \sin(t^2)), \mathrm{d}t$，$\displaystyle f(x) = \int_0^{g(x)} \frac{1}{\sqrt{1+t^3}}\,\mathrm{d}t$，求 $f'\big(\frac{\pi}{2}\big)$。

解. $\displaystyle f'(x) = \frac{1}{\sqrt{1+g(x)^3}} \cdot g'(x)$，$g'(x) = (1 + \sin(\cos^2 x)) \cdot (-\sin x)$。代入 $x = \frac{\pi}{2}$，$g\big(\frac{\pi}{2}\big) = 0$，$g'\big(\frac{\pi}{2}\big) = -1$，故 $f'\big(\frac{\pi}{2}\big) = -1$。

例. 若 $\displaystyle\int_0^{x^2} f(t)\,\mathrm{d}t = x\sin\pi x$，求 $f'(9)$。

解. $\int_0^{x^2} f(t)\,\mathrm{d}t = x\sin\pi x$ 兩邊對 x 微分得 $f(x^2)\cdot 2x = \sin\pi x + x\cdot\pi\cos\pi x$。兩邊再對 x 微分得 $(f'(x^2)\cdot 2x)\cdot 2x + f(x^2)\cdot 2 = \pi\cos\pi x + \pi\cos\pi x - x\cdot\pi^2\sin\pi x$。代入 $x=3$，則 $(f'(9)\cdot 2\cdot 3)\cdot 2\cdot 3 + f(9)\cdot 2 = \pi\cos 3\pi + \pi\cos 3\pi - 3\cdot\pi^2\sin 3\pi \implies f'(9)\cdot 36 + f(9)\cdot 2 = -2\pi$。又 $f(3^2)\cdot(2\cdot 3) = \sin 3\pi + 3\cdot\pi\cos 3\pi \implies f(9) = -\dfrac{\pi}{2}$，故 $f'(9) = -\dfrac{\pi}{36}$。

例. 求函數 f 與 $a\in\mathbb{R}$ 使 $6 + \int_a^x \dfrac{f(t)}{t^2}\,\mathrm{d}t = 2\sqrt{x},\ \forall x>0$。

解. $6 + \int_a^x \dfrac{f(t)}{t^2}\,\mathrm{d}t = 2\sqrt{x}$ 兩邊對 x 微分得 $\dfrac{f(x)}{x^2} = \dfrac{1}{\sqrt{x}} \implies f(x) = x^{\frac{3}{2}}$。代入原式得 $6 + \int_a^x \dfrac{t^{\frac{3}{2}}}{t^2}\,\mathrm{d}t = 6 + \int_a^x \dfrac{1}{\sqrt{t}}\,\mathrm{d}t = 6 + 2\sqrt{t}\Big|_a^x = 6 + 2\sqrt{x} - 2\sqrt{a} = 2\sqrt{x} \implies a = 9$。

4.3 積分技巧

三角函數代換

部份分式

4.4 瑕積分

定義 (瑕積分).

- 若 (i) f 在 I 有無限函數值或 (ii) I 為無限區間，$\int_I f(x)\,\mathrm{d}x$ 稱為瑕積分 (improper integral)。
- $\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = \lim\limits_{\substack{a\to-\infty\\ b\to\infty}} \int_a^b f(x)\,\mathrm{d}x$。

例. 求 $\int_{-\infty}^{\infty} \dfrac{1}{1+x^2}\,\mathrm{d}x$。

解. $\int_{-\infty}^{\infty} \dfrac{1}{1+x^2}\,\mathrm{d}x = 2\int_0^{\infty} \dfrac{1}{1+x^2}\,\mathrm{d}x = 2\lim\limits_{b\to\infty}\int_0^b \dfrac{1}{1+x^2}\,\mathrm{d}x = 2\lim\limits_{b\to\infty}\tan^{-1} b = 2\cdot\dfrac{\pi}{2} = \pi$。

例. 求 $\int_1^{\infty} \dfrac{1}{x^2}\,\mathrm{d}x$。

解. $\int_1^{\infty} \dfrac{1}{x^2}\,\mathrm{d}x = \lim\limits_{b\to\infty}\int_1^b \dfrac{1}{x^2}\,\mathrm{d}x = \lim\limits_{b\to\infty}\big(-\dfrac{1}{b}+1\big) = 1$

例. 求 $\int_0^1 \dfrac{1}{\sqrt{x}}\,\mathrm{d}x$。

解. $\int_0^1 \dfrac{1}{\sqrt{x}}\,\mathrm{d}x = \lim\limits_{c\to 0+}\int_c^1 \dfrac{1}{\sqrt{x}}\,\mathrm{d}x = \lim\limits_{c\to 0+} 2\sqrt{x}\Big|_c^1 = \lim\limits_{c\to 0+}(2-2\sqrt{c}) = 2$

結論. 給定 $0<a<\infty$。

- $\int_a^{\infty} \dfrac{1}{x^p}\,\mathrm{d}x$ 當 $p>1$ 收斂至 $\dfrac{a^{1-p}}{p-1}$，當 $p\leqslant 1$ 發散至 ∞。
- $\int_0^{a} \dfrac{1}{x^p}\,\mathrm{d}x$ 當 $p<1$ 收斂至 $\dfrac{a^{1-p}}{1-p}$，當 $p\geqslant 1$ 發散至 ∞。

結論. 令 $-\infty \leqslant a < b \leqslant \infty$，函數 f、g 在 (a,b) 連續，$0 \leqslant f(x) \leqslant g(x)\ \forall x$。

- 若 $\displaystyle\int_a^b g(x)\,\mathrm{d}x$ 收斂，$\displaystyle\int_a^b f(x)\,\mathrm{d}x$ 收斂。
- 若 $\displaystyle\int_a^b f(x)\,\mathrm{d}x$ 發散，$\displaystyle\int_a^b g(x)\,\mathrm{d}x$ 發散。

例. 證明 $\displaystyle\int_0^\infty e^{-x^2}\,\mathrm{d}x$ 收斂。

解. 由 $e^{-x^2} \leqslant 1\ \forall 0 \leqslant x < 1$ 及 $e^{-x^2} \leqslant e^{-x}\ \forall x \geqslant 1$，$\displaystyle\int_0^\infty e^{-x^2}\,\mathrm{d}x = \int_0^1 e^{-x^2}\,\mathrm{d}x + \int_1^\infty e^{-x^2}\,\mathrm{d}x \leqslant \int_0^1 1\,\mathrm{d}x + \int_1^\infty e^{-x}\,\mathrm{d}x = 1 + \frac{1}{e}$，故 $\displaystyle\int_0^\infty e^{-x^2}\,\mathrm{d}x$ 收斂。

例. 證明 $\forall x > 0$，$\displaystyle\Gamma(x) \equiv \int_0^\infty t^{x-1}e^{-t}\,\mathrm{d}t$ 收斂。

解. 由 $\displaystyle\lim_{t\to\infty} t^{x-1}e^{-\frac{t}{2}} = 0, \exists T > 0$ 使 $t^{x-1}e^{-\frac{t}{2}} \leqslant 1\ \forall t \geqslant T$。則 $\displaystyle\int_0^\infty t^{x-1}e^{-t}\,\mathrm{d}t = \int_0^T t^{x-1}e^{-t}\,\mathrm{d}t + \int_T^\infty t^{x-1}e^{-t}\,\mathrm{d}t \leqslant \int_0^T t^{x-1}\,\mathrm{d}t + \int_T^\infty 1\cdot e^{-\frac{t}{2}}\,\mathrm{d}t = \frac{T^x}{x} + 2e^{-\frac{T}{2}}$。故 $\forall x > 0$；$\displaystyle\int_0^\infty t^{x-1}e^{-t}\,\mathrm{d}t$ 收斂。

例. 證明 $\forall x > 0$，$\Gamma(x+1) = x\,\Gamma(x)$。

解. $\displaystyle\Gamma(x+1) = \int_0^\infty t^x e^{-t}\,\mathrm{d}t = \lim_{\substack{a\to 0+\\ b\to\infty}} \int_a^b t^x e^{-t}\,\mathrm{d}t$。令 $u = t^x$，則 $\mathrm{d}u = xt^{x-1}\,\mathrm{d}t$。令 $\mathrm{d}v = e^{-t}\,\mathrm{d}t$，則 $v = -e^{-t}$。

故 $\displaystyle\lim_{\substack{a\to 0+\\ b\to\infty}} \int_a^b t^x e^{-t}\,\mathrm{d}t = \lim_{\substack{a\to 0+\\ b\to\infty}} \Big(-t^x e^{-t}\Big|_a^b + \int_a^b e^{-t}\cdot xt^{x-1}\,\mathrm{d}t\Big) = 0 + x\int_0^\infty t^{x-1}e^{-t}\,\mathrm{d}t = x\,\Gamma(x)$。

例. 由 $\displaystyle\int_0^\infty e^{-t^2}\,\mathrm{d}t = \frac{\sqrt{\pi}}{2}$，求 $\Gamma\Big(\frac{1}{2}\Big)$ 與 $\Gamma\Big(\frac{3}{2}\Big)$。

解.

1. $\displaystyle\Gamma\Big(\frac{1}{2}\Big) = \int_0^\infty t^{-\frac{1}{2}}e^{-t}\,\mathrm{d}t = \int_0^\infty \frac{e^{-t}}{\sqrt{t}}\,\mathrm{d}t$。令 $u = \sqrt{t}$，則 $t = u^2$，$\mathrm{d}u = \dfrac{1}{2\sqrt{t}}\,\mathrm{d}t \implies \dfrac{1}{\sqrt{t}}\,\mathrm{d}t = 2\,\mathrm{d}u$。積分範圍 t 由 0 至 ∞，則變數變換後 u 由 $\sqrt{0} = 0$ 至 $\sqrt{\infty} = \infty$，故 $\displaystyle\Gamma\Big(\frac{1}{2}\Big) = \int_0^\infty \frac{e^{-t}}{\sqrt{t}}\,\mathrm{d}t = 2\int_0^\infty e^{-u^2}\,\mathrm{d}u = 2\cdot\frac{\sqrt{\pi}}{2} = \sqrt{\pi}$。
2. 由 $\Gamma(x+1) = x\,\Gamma(x)$，$\displaystyle\Gamma\Big(\frac{3}{2}\Big) = \Gamma\Big(\frac{1}{2}+1\Big) = \frac{1}{2}\,\Gamma\Big(\frac{1}{2}\Big) = \frac{1}{2}\sqrt{\pi}$。

例. 由 $\displaystyle\int_0^\infty e^{-t^2}\,\mathrm{d}t = \frac{\sqrt{\pi}}{2}$ 且 $\alpha > 0$，求 $\displaystyle\int_0^\infty t\,e^{-\alpha t^2}\,\mathrm{d}t$ 與 $\displaystyle\int_0^\infty t^2\,e^{-\alpha t^2}\,\mathrm{d}t$。

解.

1. 令 $u = \alpha t^2$，則 $\mathrm{d}u = 2\alpha t\,\mathrm{d}t \implies t\,\mathrm{d}t = \dfrac{\mathrm{d}u}{2\alpha}$。積分範圍 t 由 0 至 ∞，則變數變換後 u 由 $\alpha\cdot 0^2 = 0$ 至 $\alpha\cdot\infty^2 = \infty$，故 $\displaystyle\int_0^\infty t\,e^{-\alpha t^2}\,\mathrm{d}t = \int_0^\infty e^{-u}\,\frac{\mathrm{d}u}{2\alpha} = \frac{1}{2\alpha}\int_0^\infty e^{-u}\,\mathrm{d}u = \frac{1}{2\alpha}\Big(-e^{-u}\Big|_0^\infty\Big) = \frac{1}{2\alpha}$。
2. 令 $u = \alpha t^2$，則 $t = \dfrac{\sqrt{u}}{\sqrt{\alpha}}$，$\mathrm{d}u = 2\alpha t\,\mathrm{d}t \implies t\,\mathrm{d}t = \dfrac{\mathrm{d}u}{2\alpha}$。積分範圍 t 由 0 至 ∞，則變數變換後 u 由 $\alpha\cdot 0^2 = 0$ 至 $\alpha\cdot\infty^2 = \infty$，故 $\displaystyle\int_0^\infty t^2\,e^{-\alpha t^2}\,\mathrm{d}t = \int_0^\infty t\,e^{-\alpha t^2}\,t\,\mathrm{d}t = \int_0^\infty \frac{\sqrt{u}}{\sqrt{\alpha}}\,e^{-u}\,\frac{\mathrm{d}u}{2\alpha} = \frac{1}{2\alpha^{\frac{3}{2}}}\int_0^\infty u^{\frac{1}{2}}\,e^{-u}\,\mathrm{d}u = \frac{1}{2\alpha^{\frac{3}{2}}}\cdot\Gamma\Big(\frac{3}{2}\Big) = \frac{1}{2\alpha^{\frac{3}{2}}}\cdot\frac{1}{2}\sqrt{\pi} = \frac{\sqrt{\pi}}{4\alpha^{\frac{3}{2}}}$。